Rohan Malhotra

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EDUCATION

New York University, Courant Institute of Mathematical Sciences — New York, NY
B.A. in Mathematics and Computer Science (Double Major) — *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Probability & Statistics, Real Analysis, Linear Algebra, Numerical Methods, Differential Equations

EXPERIENCE

Kalshi — New York, NY
Quantitative Developer — *Jan 2026 – May 2026*

- Built a C++ hedging engine that converts contract prices (cents, probability, bps) into real-time exposure.
- Implemented fractional-Kelly position sizing with max-allocation caps to output recommended stake sizes.
- Co-authored research modeling job-loss risk as a macro-sensitive Poisson hazard with jump-diffusion income.
- Measured tail-risk reduction from Kalshi hedges with minimum-variance hedge ratios and Monte Carlo simulation.

IBM — Dallas, TX
Applications Developer Intern & Robotics Engineer Intern — *May 2026 – Aug 2026*

- Built an XGBoost cash-flow model, 13-day rolling cash forecasts, and SOFR loan models in Oracle EPM.
- Built a real-time YOLO11 vision stack for Boston Dynamics Spot (99.5% mAP@50) with gRPC and lock-free queues.

Hume Center for National Security and Technology, Virginia Tech — Blacksburg, VA
Aerospace Researcher — *Nov 2024 – May 2025*

- Programmed imaging and signal-processing tests in C for ContentCube, a CubeSat deployed from the ISS in 2025.
- Engineered a test environment simulating concurrent imaging workloads to resolve race conditions and memory bugs.

RESEARCH

Reddit Data in Quantitative Financial Models (published) — Jan 2025 – May 2025
Co-author, Virginia Tech (VTechWorks) — *hdl.handle.net/10919/124730*

- Co-authored a published paper on how r/wallstreetbets sentiment after the GameStop squeeze relates to volatility.
- Applied NLP sentiment scoring and regression models in Python to quantify sentiment's impact on equity returns.

An Economic Approach to Optimize Capital Allocation (published) — Aug 2024 – Dec 2024
Author, Virginia Tech (VTechWorks) — *hdl.handle.net/10919/124871*

- Applied the Kelly Criterion to capital allocation, risk of ruin, and long-run growth in investing and insurance.
- Ran Monte Carlo tests and backtests in Python comparing full vs. fractional Kelly under drawdowns and costs.

PROJECTS

Refrax | refrax.io | *C++, Next.js, TypeScript, PostgreSQL* — Jun 2025 – Feb 2026

- Designed low-latency REST/WebSocket pipelines streaming market data into a PostgreSQL research environment.
- Implemented C++ risk engines (Greeks, P&L simulation, portfolio exposure) optimized for parallel execution.

Kalshi Live Kelly Dashboard | *FastAPI, WebSockets, Plotly* — 2026

- Built a live dashboard that streams Kalshi prices over WebSockets and tests position sizes with Kelly sizing.

American Options Pricing System | *Python, NumPy, Alpaca API*

- Built a scheduled pipeline that pulls option data, prices American options with a binomial tree, and logs runs.

LEADERSHIP

PIVOT (AMP Lab) | vtpivot.org — Blacksburg, VA
Founder & Project Lead — *Feb 2024 – Dec 2025*

- Founded and led a multi-university student research group running physics and quantitative-finance projects.
- Developed a physics-based trading model with a C++ signal engine and automated execution on a Raspberry Pi.

SKILLS

Languages: C++, Python, C, SQL, Java, Rust, R, MATLAB
Quant & Math: Stochastic processes, Monte Carlo, Kelly sizing, regression, time series, options pricing
Libraries: NumPy, Pandas, SciPy, statsmodels, scikit-learn, XGBoost, PyTorch
Systems: Linux, Git, Docker, PostgreSQL, WebSockets, gRPC, multithreading, lock-free queues, AWS